

Multiplier-Free Sample-Indexed Lookup Equalization for Nonlinear Wireline SerDes Receivers

Design study, NRZ wireline link model, Lattice iCE40 HX8K target

Abstract—Receiver-side nonlinearity in high-loss wireline links cannot be undone by linear equalization, and this observation is routinely used to motivate neural-network (NN) equalizers. We show that the motivation is incomplete. If the front-end nonlinearity is memoryless, its inverse is also memoryless and is therefore exactly a 2^B -entry lookup table over the ADC code, costing no multipliers; measured on a 25 dB skin-effect channel with *tanh* compression, this 128-entry table reduces BER by 20.9× over an NLMS FFE+DFE baseline and outperforms a 96-multiplier MLP by 2.02×, after which the MLP adds nothing (0.94×). A NN earns its multipliers only when the nonlinearity has memory. We create that regime by band-limiting after compression, and identify a narrow operating window (3-dB bandwidth $\approx 0.35/T$) in which an MLP beats the memoryless table on 8/8 seeds. Within that window we prove and measure a *decision-domain ceiling*: a linear DFE, a decision-indexed ROM, and a multiplier-free Volterra expansion over ± 1 decisions all saturate near $3.1\times$ regardless of parameter count (8 to 1351 parameters), because past decisions discard amplitude. Moving the index from decisions to quantized samples removes the ceiling: a 256-entry, 1-bit-per-entry table placed after an 11-tap FFE achieves **7.98×** with *zero* back-end multipliers, exceeding a 96-multiplier MLP (5.80×) that does not fit the target device. Place-and-routed on iCE40 HX8K, the proposed equalizer costs 3 037 LC and 41.4 MHz against 2 999 LC and 34.7 MHz for the linear baseline — **+1.3 % area, 1.19× clock, 8× lower BER**. We then look for the boundary of the multiplier-free approach. Extending the model with slew-rate limiting, AC-coupling droop, far-end crosstalk, supply-gain modulation, sampling jitter and a via-stub channel resonance, we find that the binding variable is *not* the memory of the nonlinearity — ten octaves of added memory move the comparison by 15 % — but its *severity*: once both arms are given the zero-multiplier memoryless inverse, the table holds to a compression index $\kappa \approx 1.6$ and a 12-multiplier network wins at $\kappa=2.0$ on 8/8 seeds. Three multiplier-free attempts to extend the table past that point — more cells, a leaky-accumulator memory field, and a joint amplitude/decision index evaluated with ideal feedback — all fail, and we report them. Finally we bound the whole comparison: a calibrated matched-filter bound and an achievable reduced-state sequence detector show that every structure here — ours included — sits at least 25× from what the same channel and noise permit, *except* under post-compression band-limiting, where the table beats the best linear-model sequence detector tenfold. Repeating the comparison on a channel built from geometry and materials by an independent library, and on a cascade of a measured two-port, the table gains **8.48×** over the linear baseline in that regime and 1.04–1.59× outside it. Finally, synthesizing the same RTL for four device families reverses the area conclusion: on Xilinx 7-series the network costs 391 fabric cells and 12 DSP slices against 1316 cells for the table, so the multiplier-free result is bound to devices whose multipliers are themselves fabric. We give the full signal model, training algorithms, fixed-point definitions, and RTL.

Index Terms—Wireline SerDes, decision-feedback equalization, nonlinear equalization, lookup-table detection, Volterra series, quantization, FPGA implementation, crosstalk, timing jitter, channel modeling.

I. INTRODUCTION

In a high-loss wireline channel the pulse response spreads over many unit intervals and the eye closes through intersymbol interference (ISI). The standard remedy is a cascade of a continuous-time linear equalizer (CTLE), a feed-forward equalizer (FFE), and a decision-feedback equalizer (DFE). For a linear channel with additive Gaussian noise this cascade is asymptotically optimal in the MMSE sense, so any nonlinear detector can at best tie it.

Real receivers are not linear. Gain compression in the analog front end, buffer saturation, and driver nonlinearity all warp the signal *after* ISI has spread it. Such a warp has no linear inverse, and this is the standard argument for a neural-network equalizer.

The argument has a gap. “A linear FFE/DFE cannot invert it” and “a neural network is required” are different claims, and between them lies the entire class of *nonlinear but non-neural* correctors. In this paper we close that gap and find that it dominates the design space we studied.

Our contributions are:

- A proposition and measurement showing that a *memoryless* front-end nonlinearity admits an exact 2^B -entry lookup inverse over the ADC code, with zero multipliers, which outperforms an MLP of 96 multipliers by 2.02× and renders it redundant (Sec. V).

- Identification of the regime in which a NN does earn its cost — nonlinearity *with memory*, created by post-compression band-limiting — and measurement of how narrow that window is (Sec. VI).
- A *decision-domain ceiling*: an information argument plus measurement showing that a linear DFE, a full decision-indexed ROM, and a multiplier-free ± 1 Volterra expansion all saturate near $3.1\times$ irrespective of parameter count (Sec. VII). We prove that any polynomial in ± 1 decision variables is realizable with sign-controlled additions only.
- A *sample-indexed lookup equalizer* that removes the ceiling at zero back-end multiplier cost, with a hardware-realizable shift-and-clamp quantizer, its training rule, complexity analysis, RTL, and place-and-route results (Sec. VIII–X).

II. SYSTEM MODEL

A. Channel

NRZ symbols $b[n] \in \{-1, +1\}$ at rate $1/T$ are oversampled by M samples per symbol. The channel magnitude follows a skin-effect law,

$$|H_{\text{ch}}(f)| = 10^{-(L_{\text{dB}}/20) \sqrt{f/f_N}}, \quad f_N = 1/(2T) \quad (1)$$

with L_{dB} the Nyquist loss. Phase is not free: a linear-phase reconstruction produces a symmetric cursor set, which removes the very post-cursor asymmetry a DFE exists to

cancel. We therefore build the minimum-phase response from the real cepstrum. With N -point DFTs and $c = \text{IDFT}\{\ln|H|\}$,

$$m[k] = c[0], 2c[k]_{0 < k < N/2}, c[N/2], 0 \quad (2)$$

$$h = \text{IDFT}\{\exp(\text{DFT}\{m\})\} \quad (3)$$

Connector and via-stub reflections are applied multiplicatively,

$$H(f) = H_{\text{ch}}(f) \cdot (1 + \sum_k \Gamma_k e^{-j2\pi f \tau_k}) \quad (4)$$

which places a notch wherever $f\tau = 1/2$, of depth $20 \log_{10}(1 - |\Gamma|)$. Unlike skin-effect loss, a notch removes one frequency and cannot be filled by the smooth boost of a CTLE. In the time domain it appears as an isolated cursor at lag τ/T .

B. Noise and receiver nonlinearity

Additive Gaussian noise is referenced to the main cursor of a lossless link, so that the loss-free, unequalized link satisfies a closed form used later for calibration:

$$r(t) = (x * h)(t) + w(t), \quad \sigma = 10^{-\text{SNR}_{\text{dB}}/20} \quad (5)$$

Front-end compression is applied *after* noise and *before* equalization, which is where a VGA/CTLE stage sits:

$$g(u) = \frac{\tanh(a u)}{a}, \quad a = \frac{\kappa}{\text{RMS}(u)} \quad (6)$$

where κ is the compression strength. Two properties of (6) matter. First, it is **memoryless**: g acts on one sample. Second, applying it at the transmitter would do nothing at all — NRZ has only two levels, so a memoryless map is a pure gain change and the slicer output is unchanged. Nonlinearity acquires meaning only after ISI has spread the levels, which is why it is placed at the receiver.

C. Post-compression band-limiting

A real receiver filters again after the compressing stage — CTLE loading, ADC input network, package. We model this with a Hann low-pass q of 3-dB bandwidth β/T :

$$v[n] = \sum_m q[m] \cdot g(r[n-m]) \quad (7)$$

Equation (7) is the crux of this paper. When $q = \delta$, $v[n]$ is a function of $r[n]$ alone and the nonlinearity remains memoryless. When q has support greater than one, $v[n]$ mixes several *compressed* values and **no memoryless map can recover** $r[n]$. The parameter β therefore moves the impairment continuously between two qualitatively different classes.

D. Quantization and gain control

Sampling at the pulse-response peak phase, a B -bit ADC with full scale $\rho\sigma_v$ yields

$$y[n] = \Delta \cdot \text{clip}(\text{round}(v[n]/\Delta), -2^{B-1}, 2^{B-1}-1) \quad (8)$$

with $\Delta = \rho \sigma_v / 2^{B-1}$, σ_v estimated over the training preamble. A digital AGC then normalizes by the measured main-cursor gain $y = E[y b]$ taken over the preamble only.

E. Calibration

With $L_{\text{dB}} = 0$, $\kappa = 0$ and no equalizer, the model must reproduce $\text{BER} = Q(10^{\text{SNR}/20})$. Over 4×10^5 bits the measured values agree to 0.49 %, 3.03 % and 1.30 % at 6, 8 and 10 dB, all inside the 3σ counting error. An SNR without such a definition is an arbitrary number.

III. BASELINE EQUALIZERS

A. NLMS feed-forward equalizer

The FFE is adapted by normalized LMS over the preamble. With tap vector $\mathbf{w} \in \mathbb{R}^L$, input window $\mathbf{y}[n] = (y[n], \dots, y[n-L+1])^T$ and reference delay $D = \lfloor L/2 \rfloor$,

$$e[n] = \mathbf{w}^T \mathbf{y}[n] - b[n-D] \quad (9)$$

$$\mathbf{w} \leftarrow \mathbf{w} - \frac{\mu}{L P} e[n] \mathbf{y}[n], \quad P = E[y^2] \quad (10)$$

The division by L is not cosmetic. The LMS stability bound is $\mu < 2/(L \sigma_y^2)$, so normalizing by power alone diverges as taps are added; with power-only normalization a 29-tap FFE diverged to tap magnitudes of 1.08×10^6 in our runs, which would have been misread as “more taps do not help.” We use $\mu = 0.2$ throughout.

B. Decision feedback with arbitrary tap placement

The slicer feeds back its *own* decisions — substituting true bits removes error propagation and inflates BER by orders of magnitude. For tap positions $p_1 \dots p_K$ (not necessarily contiguous) and coefficients c_k ,

$$d[n] = \text{sgn}(y_f[n] - \sum_{k=1}^K c_k d[n-p_k]) \quad (11)$$

Because $d \in \{-1, +1\}$, every product in (11) is a sign selection: **a DFE tap contains no multiplier**, only an add/subtract. This is the single most important cost asymmetry in the paper and it recurs in Sec. VII.

Free tap placement matters more than tap count. For $\Gamma = 0.35$ at $\tau = 11T$, the residual cursor after an 11-tap FFE is ≤ 0.055 at lags 1–10 and **+0.358** at lag 11. A single floating tap at 11 is worth $8.2\times$ in BER over a tap at 10 or 12; a *pair* of adjacent taps removes that cliff entirely (Table I). Locating the reflection by correlation needs only 200 symbols (12/12 seeds), so tracking a drifting reflection is an operational matter, not a design constraint.

TABLE I

Cost of misplacing a floating DFE tap. $\Gamma=0.35$ at $\tau=11T$, 11-tap FFE, 5×10^5 bits, 4 seeds.

Floating tap position	BER	vs. correct
{11} — correct	1.83e-03	1.0×
{9}, {10}, {12}, {13}	1.49–1.52e-02	8.2–8.3×
{10,11}, {11,12}, {10,11,12}	1.74–1.79e-03	1.0×

C. Word length

Coefficients and ADC are quantized jointly, since they are not chosen independently in practice. Sweeping channel loss with

each point re-normalized to a BER of 10^{-3} , 7 bits is flat to 20 dB; at 25 dB the 6-bit setting averages +26.6 % but spreads over [+6.5, +69.6] % across seeds, which is why it is not usable. Above 30 dB no width in the swept range satisfies both a small mean and a small spread. The ADC optimum is $\rho = 2.5\sigma$ at every resolution, with a clipping fraction of 0.43 % rather than zero — a well-set ADC deliberately clips a little. The joint operating point used throughout is **7-bit coefficients, 7-bit ADC**, which costs +2.2 % BER relative to floating point.

IV. NEURAL-NETWORK EQUALIZER

A. Architecture and training

The NN equalizer is a single-hidden-layer MLP over a centred tapped-delay line of the ADC samples. With window half-width A , $L = 2A+1$, hidden width H :

$$\mathbf{x}[n] = (y[n-A], \dots, y[n+A])^T \quad (12)$$

$$\mathbf{z} = \mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1, \quad \mathbf{h} = \varphi(\mathbf{z}), \quad u = \mathbf{w}_2^T \mathbf{h} + b_2 \quad (13)$$

$$d[n] = \text{sgn}(u[n]) \quad (14)$$

Training minimizes the MSE to $b[n]$ with Adam over the first 30 % of the record; BER is counted on the remainder only. The multiplier count is $LH + H$.

The activation is chosen by BER, not by approximation error. Hardware cannot evaluate $\tanh$ directly, so we compare $\tanh$ against $\text{hardtanh}(z) = \text{clip}(z, -1, 1)$ — two comparisons, no table — and a three-segment piecewise-linear fit *pwl3*. At 12 multipliers *hardtanh* wins on 7 of 8 seeds (sign test $p \approx 0.04$). Notably *pwl3* approximates $\tanh$ far better yet is worst in BER at every size (1.4–1.9×), because the network adapts to whatever nonlinearity it is given: **a better approximation is not a better equalizer**.

B. Quantization

Per-tensor scaling destroys the network. A trained $\mathbf{W}_1$ of shape 21×16 has max 2.4003 and median 0.1133 — a spread of $21 \times$ — so a 4-bit tensor-wide step of $\max/7$ rounds all but 138 of 369 weights to zero. Per-output-channel scaling,

$$s_j = \max_i |W_{ij}|, \quad W_{ij} = \frac{s_j}{S} \text{clip}(\text{round}(S W_{ij}/s_j), -S-1, S) \quad (15)$$

with $S = 2^{B_w-1}-1$, keeps 282 of 369 alive at 4 bits. Linear equalizer taps span one decade and tolerate tensor scaling; NN weights do not. Quantization-aware training with a straight-through estimator and a grid frozen at the float solution was implemented but **did not establish a gain** at this scale: the winning width moved between variants and the seed spread exceeded the differences. What is established is post-training quantization with per-channel scaling.

The dominant lightweighting axis in the swept space is network size, not width:

TABLE II

Network size versus gain over the 8-bit linear baseline. $\kappa=1.0$, 25 dB, SNR 30 dB, 4 seeds.

Window	Hidden	Mults	BER	Gain	Gain/mult
5	2	12	7.671e-03	3.1×	2.89
5	4	24	4.775e-03	5.1×	2.32
9	8	80	3.409e-03	7.1×	0.97
21	16	352	3.161e-03	7.6×	0.24

Twelve multipliers buy 41 % of the available gain; a further $15 \times$ in multipliers buys $1.5 \times$ more. By comparison, $8 \rightarrow 6$ bit saves 14.8 % area for 31 % worse BER — a far smaller handle.

C. Fixed point

The integer reference that the RTL must match is

$$\mathbf{z}^q = (\mathbf{X}^q \mathbf{W}_1^q + \mathbf{b}_1^q 2^F) \gg F \quad (16)$$

$$\mathbf{h}^q = \text{clip}(\mathbf{z}^q, -2^F, 2^F), \quad u^q = \mathbf{h}^q \mathbf{w}_2^q + b_2^q 2^F \quad (17)$$

where F is the fractional width and $\gg$ is an arithmetic shift. Sweeping total width and F independently separates two effects that are usually conflated. **Performance is set by the integer headroom, not the fractional resolution:** width $8/F = 6$ and width $10/F = 8$ — both with one integer bit — are equally bad ($1.84 \times$ and $1.86 \times$ float), while any format with two integer bits and $F \geq 4$ matches float (0.98 – $1.00 \times$). The minimum is therefore **Q2.4 in 7 bits**, which is the same width the linear FFE needs. Choosing Q3.6/10-bit instead, as we did initially, costs $1.9 \times$ the area for identical BER.

Accumulator widths are sized so that no truncation occurs: with W_x , W_w the sample and weight widths,

$$W_z = W_x + W_w + \lceil \log_2(L+1) \rceil, \quad W_h = F + 2 \quad (18)$$

$$W_u = W_h + W_w + \lceil \log_2(H+1) \rceil \quad (19)$$

V. THE MEMORYLESS INVERSE

We now supply the control that the usual motivation omits.

PROPOSITION 1. *Let the receiver impairment be $y[n] = Q_B(g(r[n]))$ with g memoryless and strictly monotone and Q_B a B -bit quantizer. Then the MMSE memoryless corrector is exactly the table*

$$T[c] = E[r[n] | y[n] = c], \quad c \in \{0, \dots, 2^B - 1\} \quad (20)$$

and no memoryless corrector of any complexity can do better.

Proof. A memoryless corrector is a map from the observation alphabet to $\mathbb{R}$. The observation alphabet is the set of 2^B ADC codes, hence any such map is a table of 2^B entries. Among all such maps the conditional mean minimizes mean-square error. $\square$

The consequence is a hardware statement: **inverting a memoryless nonlinearity costs one ROM read and no multiplier**. For $B = 7$ that is 128 entries. When the compression law is known the table can also be written in closed form,

$$T_{\text{inv}}(z) = \frac{1}{a} \operatorname{artanh}(\operatorname{clip}(az, -1+\varepsilon, 1-\varepsilon)) \quad (21)$$

which is an *upper bound* on any data-driven memoryless corrector, since it assumes exact knowledge of a . The clipping in (21) is physical, not numerical: artanh diverges as $|az| \rightarrow 1$, and that divergence is the statement that information destroyed in saturation cannot be recovered — inverting there only amplifies noise.

TABLE III

The missing control. 25 dB, SNR 30 dB, $\kappa=1.0$, no post-filter, 3×10^5 bits, 6 seeds.

Equalizer	BER	Gain	Mults	Entries
Linear FFE 11 + DFE 8	2.3379e-02	1.00×	11	—
Inverse table + linear	1.1190e-03	20.9×	11	128
MLP, 96 multipliers	2.2563e-03	10.4×	96	—
MLP, 12 multipliers	4.3746e-03	5.3×	12	—
Inverse table + MLP 96	1.1937e-03	19.6×	96	128

Table III settles the question for this impairment. The 128-entry table beats the 96-multiplier network by 2.02× and the 12-multiplier network by 3.9×, adding no multipliers. More decisively, **once the table is applied the network contributes nothing** (0.94×; 3.3×10^6 bits would be needed to separate the two). The residual impairment after memoryless correction is linear, and a linear equalizer handles it.

The precise claim is therefore not “a neural network solves what a linear equalizer cannot,” but rather: *a linear FFE/DFE cannot invert a memoryless warp, and a network does so more expensively and less well than a table.*

VI. NONLINEARITY WITH MEMORY

Proposition 1 depends on the impairment being a function of a single sample. Equation (7) breaks that condition: after band-limiting, $v[n]$ depends on $g(r[n])$, $g(r[n-1])$, ... jointly, so the map $r[n] \mapsto v[n]$ is not injective and (20) is no longer sufficient. This is where a nonlinear detector with memory becomes necessary.

Sweeping β and comparing *per-seed ratios* (means of heavy-tailed BERs are dominated by a single bad seed) locates the regime:

TABLE IV

Where a network beats the memoryless table. 25 dB, SNR 30 dB, $\kappa=1.0$, 3×10^5 bits, 8 seeds.

Post-comp. BW β	MLP / table	Spread	MLP wins
∞ (no filter)	0.51×	[0.32, 0.70]	0/8
0.50	0.30×	[0.11, 0.51]	0/8
0.35	3.20×	[2.49, 5.16]	8/8
0.25	2.28×	[0.69, 11.90]	1/8
0.20	0.87×	[0.66, 1.08]	1/8

The window is narrow and one-sided in both directions: wide bandwidth leaves the impairment effectively memoryless, where the table wins; very narrow bandwidth destroys the link for every structure. At $\beta = 0.25$ the mean ratio of 2.28× is produced by a single seed at 11.90× while the network wins

only 1 of 8 — not an established result. **All experiments in Sec. VII–X are conducted at $\beta = 0.35$.**

VII. THE DECISION-DOMAIN CEILING

Within the window, what can be achieved without adding multipliers? The cost structure measured on the target device is:

TABLE V

Impairment class, matched corrector, multiplier cost.

Impairment	Matched corrector	Mults
Memoryless nonlinearity	ADC-code table, 2^B	0
Linear pre-cursor ISI	FFE	1/tap
Linear post-cursor ISI	DFE	0
Decision-dependent nonlinearity	Decision-indexed table, 2^L	0
Sample-dependent nonlinearity	MLP / Volterra on samples	many

A. Decision-indexed ROM

A linear DFE computes a *linear* function of the past L decisions. Replacing it by an arbitrary function of the same inputs is one table of 2^L entries and still contains no multiplier. With index

$$t[n] = \sum_{k=0}^{L-1} 2^k \cdot \mathbf{1}[d[n-1-k] > 0] \quad (22)$$

$$T[i] = \mathbb{E}[y_t[n] - b[n] \mid t[n]=i] \quad (23)$$

$$d[n] = \operatorname{sgn}(y_t[n] - T[t[n]]) \quad (24)$$

The table is trained on the preamble with known bits — the standard situation — but indexed at run time by the receiver's own decisions, so error propagation behaves exactly as in a DFE. Cells with fewer than a threshold of training samples fall back to the linear DFE value; otherwise rare patterns memorize noise.

This structure strictly generalizes the DFE, so a control is available: on a *linear* channel the optimal post-cursor canceller is linear and the ROM must not win. It does not — 6.5060e-04 versus 6.5595e-04, winning 3 of 8 seeds, requiring 3.6×10^8 bits to separate. Inside the window it reaches 3.08×, but depth does not help beyond 8–10 and then hurts (2.13× at depth 12), because 4096 cells over 9×10^4 training symbols is 22 samples per cell. **The ceiling of a table is set by training data, not by area.**

B. Volterra over ± 1 decisions

PROPOSITION 2. Let $d_1 \dots d_L \in \{-1, +1\}$. Every monomial $\prod_{k \in S} d_k$ takes values in $\{-1, +1\}$, and its value is determined by the parity (XOR) of the sign bits of $\{d_k\}_{k \in S}$. Consequently any polynomial

$$f(\mathbf{d}) = w_0 + \sum_{|S| \leq D} w_S \prod_{k \in S} d_k \quad (25)$$

is realizable with sign-controlled additions only, using no multipliers, with $\sum_{j \leq D} C(L, j)$ coefficients.

Proposition 2 gives exactly the parameter sharing a table lacks: the full ROM is the degree- L case with 2^L coefficients, while truncating to degree D reduces this to $O(L^D)$ at unchanged

multiplier cost. Coefficients are fit by least squares on the preamble design matrix of monomials; for evaluation we expand them back into a 2^L table, the two realizations being equivalent in hardware (one adder tree of $|S|$ terms, or one ROM read).

It does not work.

TABLE VI

Decision-domain structures at $\beta=0.35$. All win over linear on 8/8 seeds; none escapes $\sim 3.1\times$.

Structure	Params	Gain	Spread	Mults
Linear DFE, 8 taps	8	1.00×	—	0
Volterra $L=12, D=2$	79	1.04×	[0.98, 1.11]	0
Volterra $L=16, D=2$	137	1.10×	[1.00, 1.22]	0
Volterra $L=12, D=3$	299	2.68×	[1.99, 3.17]	0
Volterra $L=16, D=3$	697	2.87×	[2.15, 3.38]	0
Volterra $L=20, D=3$	1351	2.91×	[2.25, 3.47]	0
Decision ROM, $L=8$	256	3.08×	[2.18, 3.64]	0
Decision ROM, $L=10$	1024	3.09×	[2.11, 3.91]	0
Decision ROM, $L=12$	4094	2.13×	[1.69, 2.55]	0
MLP, 96 multipliers	~ 193	5.80×	[4.81, 6.71]	96

C. Diagnosis

Three structurally different correctors — linear, degree-truncated polynomial, and full table — saturate at the same value while their parameter counts span 8 to 1351. An MLP with roughly 193 parameters reaches $5.80\times$. The common property of the saturating three is their *observation*.

PROPOSITION 3. *Let O_n be the observation available to a detector. The minimum achievable error probability given O_n is that of the MAP rule $\text{sgn } E[b[n] | O_n]$, and a lookup table indexed by a finite quantization of O_n converges to it as the training record grows. Hence detectors that differ only in O_n are ordered by the information O_n carries about $b[n]$.*

Decisions $d[n-1], \dots, d[n-L]$ are one-bit functions of *past* observations. They carry no information about the amplitude of the current sample, and it is precisely amplitude that (7) mixes. **The limit is the observation, not the parameter count** — which is why adding 1343 parameters moves the result by $0.23\times$ while changing the observation moves it by more than $2\times$.

A corollary, also measured: composing two individually correct correctors can be worse than either. Inverse table followed by decision ROM gives $2.11\times$ against $3.08\times$ for the ROM alone (7/8 seeds worse), because artanh amplifies noise in the saturated tail and the ROM then averages a noisier signal.

VIII. PROPOSED: SAMPLE-INDEXED LOOKUP EQUALIZER

Proposition 3 prescribes the fix: move the index from decisions to samples. It does *not* prescribe multipliers — a coarsely quantized sample is an address.

A. Architecture

The FFE is retained, because the table cannot perform linear equalization (Sec. X). Its accumulator $a[n]$ is quantized by an

arithmetic shift and a clamp, and W consecutive codes form the address:

$$c[n] = \text{clip}(a[n] \gg s) + 2^{Q-1}, 0, 2^{Q-1}) \quad (26)$$

$$t[n] = \sum_{k=0}^{W-1} c[n-k] \cdot 2^{Qk} \quad (27)$$

$$T[i] = E[b[n] | t[n]=i], \quad d[n] = \text{sgn}(T[t[n]]) \quad (28)$$

By Proposition 3, (28) is the empirical MAP detector on the quantized W -sample observation. Because only the sign of T is used, the stored table is **one bit per entry**: 2^{QW} bits, i.e. 256 bits at $Q=4, W=2$. The back end contains no multiplier and no adder — a shift, a clamp, a concatenation and a ROM read. There is also **no decision feedback**: past *codes* are retained, not past decisions, so there is no error propagation and no feedback loop to constrain pipelining.

The quantizer must be hardware-realizable. Stretching by a training percentile requires a multiply; (26) uses a power-of-two step chosen once from the preamble RMS. Measured, the restriction is not a cost at all: $7.98\times$ for the shift quantizer against $7.80\times$ for the percentile version.

ALGORITHM 1: TRAINING THE SAMPLE-INDEXED TABLE

1. Adapt FFE taps $\mathbf{w}$ by NLMS (9)–(10) on the preamble.
2. Estimate $\sigma = \text{RMS}(a[n])$ over the preamble; set $s = \text{round}(\log_2(\rho\sigma / 2^{Q-1}))$.
3. Zero accumulators: $S[i] \leftarrow 0, N[i] \leftarrow 0$ for all i .
4. For each preamble index n : form $c[n]$ by (26) and $t[n]$ by (27); $S[t] \leftarrow S[t] + b[n]$; $N[t] \leftarrow N[t] + 1$.
5. For each cell i : $T[i] \leftarrow 1$ if $N[i] \geq N_{\min}$ and $S[i] \geq 0$; $T[i] \leftarrow 0$ if $N[i] \geq N_{\min}$ and $S[i] < 0$; else default.
6. Write T into the ROM through the coefficient port.

Step 5 is a single pass of counting; no gradient, no iteration, and the convergence criterion is simply how many samples each cell has received. Complexity is $O(N_{\text{pre}})$ time and $O(2^{QW})$ memory.

LISTING 1: TABLE CONSTRUCTION (PYTHON)

```
def build_table(a, b, n_tr, w=2, Q=4, rho=2.5, n_min=4):
    """a: FFE accumulator; b: +-1 bits; n_tr: preamble."""
    rms = np.sqrt(np.mean(a[:n_tr]**2)) or 1.0
    step = 2.0 ** np.round(
        np.log2(rho*rms / (1 << (Q-1)))) # eq.(26)
    code = np.clip(np.floor(a / step) + (1 << (Q - 1)),
        0, (1 << Q) - 1).astype(np.int64)

    idx = np.zeros(len(a), dtype=np.int64) # eq.(27)
    for k in range(w):
        past = np.concatenate(
            [np.zeros(k, np.int64), code[:len(a) - k]])
        idx += past << (Q * k)

    cells = 1 << (Q * w)
    s = np.zeros(cells); n = np.zeros(cells)
    np.add.at(s, idx[:n_tr], b[:n_tr]) # preamble only
    np.add.at(n, idx[:n_tr], 1.0)
    T = np.where(n >= n_min, s / np.maximum(n, 1.0), 0.0)
    return np.where(T[idx] >= 0, 1.0, -1.0), T, step
```

B. Complexity

Let L_f be the FFE length. Per symbol the proposed equalizer costs L_f multiplies (FFE), one shift, two comparisons, one concatenation and one ROM access. The DFE it replaces costs K add/subtracts in a serial chain; the MLP it replaces costs $LH+H$ multiplies in two serial layers. Memory is 2^{QW} bits against KW_c bits for the DFE and $(LH+2H+1)W_w$ for the MLP.

The exponential memory in QW is the binding constraint, and it binds through data rather than area: the number of cells that receive enough training samples grows far more slowly than the number of cells. At $Q=4$, $W=2$ (256 cells) 217 are populated and the gain is 7.98×; at $W=3$ (4096 cells) only 1531 are populated and the gain collapses to 2.01×. **Enlarging the observation window without enlarging the preamble is counter-productive.**

IX. RTL AND IMPLEMENTATION

Listing 2 gives the synthesizable core. The FFE products are formed at exact width, the accumulator is sized by (18), and the quantizer is a shift with saturation. The coefficient and table write ports keep the adaptation engine out of the datapath; exporting coefficients to top-level pins instead would exhaust the device I/O long before its logic.

LISTING 2: SAMPLE-INDEXED LOOKUP EQUALIZER (VERILOG)

```
module ffe_tbl #(parameter L=11, W=7, DW=7, FW=18,
                SH=6, QB=4, WIN=2, TA=8, NT=256)(
    input wire clk, rst_n,
    input wire signed [DW-1:0] x,
    input wire cw_we, input wire [3:0] cw_addr,
    input wire signed [W-1:0] cw_data,
    input wire tw_we, input wire [TA-1:0] tw_addr,
    input wire tw_data,
    output reg d);

    reg signed [DW-1:0] sr [0:L-1];
    reg signed [W-1:0] coef [0:L-1];
    reg tbl [0:NT-1]; // 1 bit / entry

    wire signed [W+DW-1:0] prod [0:L-1];
    wire signed [FW-1:0] ext [0:L-1];
    genvar i;
    generate for (i=0;i<L;i=i+1) begin : taps
        assign prod[i] = coef[i] * sr[i];
        assign ext[i] = {(FW-W-DW){prod[i][W+DW-1]}},
                        prod[i];
    end endgenerate

    integer k; reg signed [FW-1:0] accf;
    always @* begin
        accf = 0;
        for (k=0;k<L;k=k+1) accf = accf + ext[k];
    end

    // eq.(26): arithmetic shift + clamp, no multiply
    wire signed [FW-1:0] sh = accf >> SH;
    localparam HI = (1<<(QB-1)) - 1;
    localparam LO = -(1<<(QB-1));
    wire [QB-1:0] qcode = (sh > HI) ? {QB{1'b1}} :
                          (sh < LO) ? {QB{1'b0}} :
                          (sh[QB-1:0] + {1'b1,{(QB-1){1'b0}}});

    reg [QB-1:0] qreg; // pipeline reg
    reg [QB-1:0] pcode [0:WIN-2];
    reg [TA-1:0] addr;
    always @* begin // eq.(27)
        addr = {(TA-QB){1'b0}}, qreg;
        for (k=0;k<WIN-1;k=k+1)
            addr = addr | ({(TA-QB){1'b0}}, pcode[k]
                          << (QB*(k+1)));
    end

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            for (k=0;k<L;k=k+1) begin
                sr[k] <= 0; coef[k] <= 0;
            end
            for (k=0;k<WIN-1;k=k+1) pcode[k] <= 0;
            for (k=0;k<NT;k=k+1) tbl[k] <= 1'b0;
            qreg <= 0; d <= 1'b0;
        end else begin
            if (cw_we) coef[cw_addr] <= cw_data;
            if (tw_we) tbl[tw_addr] <= tw_data;
            sr[0] <= x;
            for (k=1;k<L;k=k+1) sr[k] <= sr[k-1];
            qreg <= qcode;
            pcode[0] <= qreg;
            for (k=1;k<WIN-1;k=k+1) pcode[k] <= pcode[k-1];
            d <= tbl[addr]; // eq.(28)
        end
    end
endmodule
```

Because the structure has no feedback, the single register on *qreg* splits the critical path into *multiply* → *adder* → *shift/clamp* and *ROM read*, raising $F_{\max}$ from 26.62 to 41.38 MHz at lower area. Latency rises by one cycle; throughput is unchanged, since the equalizer emits one decision per symbol. The same cut cannot be applied

inside a DFE loop, where inserting a register delays the feedback itself.

X. RESULTS

A. Bit error rate

TABLE VII

Sample-indexed table at $\beta=0.35$. 25 dB, SNR 30 dB, $\kappa=1.0$, 3×10^5 bits, 8 seeds.

Configuration	Cells	Filled	Gain	Wins
FFE 11 + table 2×4 b	256	217	7.98×	8/8
FFE 11 + table 2×5 b	1024	796	6.86×	8/8
FFE 11 + table 3×4 b	4096	1531	2.01×	8/8
Table 2×4 b, no FFE	256	136	0.26×	0/6
Table 3×5 b, no FFE	32768	3383	0.63×	0/6

Two facts bound the claim. The 256-cell table exceeds the 96-multiplier MLP (5.80×, Table VI) using no back-end multipliers. And **without the FFE the table fails outright** (0.26–0.63×): it cannot perform linear equalization, so “multiplier-free” describes only the stage that replaces the DFE or the network, not the receiver.

TABLE IX

Place-and-route on iCE40 HX8K, 50 MHz target. BER gains at $\beta=0.35$ relative to the linear baseline.

Design	Mults	LC	$F_{\max}$	50 MHz	BER gain
FFE 11 tap, W7/DW7 (alone)	11	2237	52.38 MHz	pass	—
Linear FFE 11 + DFE 8, pipelined	11	2999	34.73 MHz	fail	1.00×
MLP 5 → 2, Q2.4, pipelined	12	2336	47.87 MHz	fail	2.55×
FFE 11 + table 2×4, unpipelined	11	3047	26.62 MHz	fail	7.98×
FFE 11 + table 2×4, pipelined	11	3037	41.38 MHz	fail	7.98×
FFE 11 + table 2×5, pipelined	11	5497	—	—	6.86×
MLP, window 11, hidden 8	96	~19000	—	does not fit	5.80×

Against the linear baseline the proposed equalizer costs **+1.3 % area**, runs **1.19× faster**, and lowers BER by **8×**. Against an area-matched network it is 30 % larger and 14 % slower but 3.1× better in BER; the network configuration that would reach 5.80× requires roughly 19 000 LC and does not fit the device.

Two scaling laws were measured. First, the table stage is **linear in cell count at ≈3.2 LC per cell** (810 LC at 256 cells, 3283 at 1024, 12698 at 4096). The stored data is only 256 bits; the area is the address decode and the write port. A 256-cell table is therefore comparable in area to an 8-tap DFE (762 LC): **multiplier-free is not free**. Second, since gain *falls* with cell count beyond 256 (Table VII) while area rises linearly, this axis offers nothing to trade.

Multiplier width follows an approximately quadratic law in the neural path: widening Q2.4/7-bit to Q2.5/8-bit costs 1.272× the area against a quadratic prediction of 1.306×, and 8 → 10 bit costs 1.494× against 1.563×. With three points we report consistency with quadratic scaling rather than fitting an exponent.

The required control is that on a linear channel, where the optimal equalizer is linear, the table must not win:

TABLE VIII

Control. Gain of FFE 11 + table 2×4 over FFE 11 + DFE 8, 6 seeds.

Condition	Gain	Spread	Wins
$\kappa=0$, no post-filter	1.14×	[0.87, 1.45]	4/6
$\kappa=0$, $\beta=0.35$	0.27×	[0.16, 0.40]	0/6
$\kappa=1.0$, no post-filter	6.36×	[2.30, 11.94]	6/6
$\kappa=1.0$, $\beta=0.35$	7.49×	[4.13, 12.21]	6/6

On a linear channel the table ties or loses badly — it observes two samples and cannot match the span of an 11+8-tap linear equalizer. This is the expected and required behaviour.

B. Area and timing

All designs were synthesized with Yosys and placed and routed with nextpnr on a Lattice iCE40 HX8K (7680 logic cells), targeting 50 MHz; no timing-failure override was used. The device has no hard multipliers, so all products map to LUTs.

C. Pipelining

One register stage raises $F_{\max}$ by 53–70 % across all three designs at no area cost (−1.4 % to +2.2 %), and every pipelined variant emits the identical decision sequence one cycle later. It does not, however, reorder the designs: the linear baseline improves by the same proportion, because only the DFE feedback loop resists cutting and that loop is not the critical path. None of the three reaches the 50 MHz target; the best is 4 % short.

XI. WHERE THE LOOKUP EQUALIZER FAILS

The previous edition of this work closed with an admission: the address space 2^{QW} must eventually grow out of reach, “*but we did not construct such a case, and therefore we have not demonstrated a regime that requires a neural equalizer.*” This section constructs the case. The outcome is not the one we predicted, and we report the prediction and its refutation together, because the refutation is the result.

A. Extended impairment set

Five impairments are added to the model of Sec. II. *Slew-rate limiting*, the finite current of an amplifier driving a capacitive load, is a nonlinear recursion whose memory grows as the limit tightens:

$$v[n] = v[n-1] + \text{clip}(u[n] - v[n-1], -S, +S) \quad (29)$$

AC-coupling droop is a first-order high-pass whose memory is set directly by a time constant, and whose interaction with a following compressor is the mechanism we intended to exploit: the amount by which a sample is compressed comes to depend on a slow baseline:

$$b[n] = a b[n-1] + (1-a)u[n], \quad v[n] = u[n] - b[n], \quad a = e^{-(30)}$$

Far-end crosstalk adds an independent PRBS aggressor through the same channel with a differentiating coupling; supply noise modulates the front-end gain multiplicatively, $y[n](1 + m \sin 2\pi n/MP)$; and residual timing error displaces the sampling instants, the samples being taken by linear interpolation:

$$t[n] = \varphi + nM + M(\sigma_{ij}g[n] + A_{sj} \sin 2\pi n/P_{sj}) \quad (31)$$

Finally the channel itself is re-parameterized. A single $\sqrt{f}$ term is the shape of conductor loss alone; a printed-circuit link also carries dielectric loss proportional to f , and the most conspicuous feature of a measured S_{21} is neither — it is the resonance of an unused via stub, which shorts the line at its quarter-wave frequency:

$$|H(f)| = 10^{-(\alpha_c \sqrt{f/f_N} + \alpha_d(f/f_N))/20} \cdot \left| 2 / (2 + j s \tan 2\pi f \tau_s) \right|^2 \quad (32)$$

Minimum phase is imposed as in (2). Two checks pin the parameterization: with $\alpha_d=0$ and $s=0$, (32) reproduces the channel of Sec. II to 4×10^{-17} ; and the notch falls at $1/(4\tau_s)$, measured at $1.00 f_N$ for $\tau_s=0.5 T$ and $0.50 f_N$ for $\tau_s=1.0 T$. At equal 20 dB Nyquist loss the two mechanisms are not interchangeable: the main cursor is 0.238 under conductor loss and 0.358 under dielectric loss.

B. Long memory does not break the table

The droop of (30) reaches 18.8 UI — an order of magnitude beyond the $W=2$ window — and the prediction was that the table would collapse there. It does not. Holding every other parameter at the operating point of Sec. X and changing one variable at a time:

TABLE X

One variable at a time from 25 dB, SNR 30 dB, $\kappa=1.0$, no post-filter, 6.4×10^5 bits, 8 seeds. Ratio is NN 5 → 2 (7-bit, 12 multipliers) over FFE 11 + table 2×4, per-seed median.

Change	NN / table	NN wins
none (reference)	1.28×	5/8
droop $\tau = 18.8$ UI	1.47×	7/8
post-comp. $\beta = 0.35$	0.56×	1/8
compression $\kappa = 1.3$	2.17×	8/8
compression $\kappa = 1.6$	2.50×	8/8
compression $\kappa = 2.0$	3.40×	8/8
loss 25 → 20 dB	1.83×	7/8
SNR 30 → 26 dB	1.23×	7/8

Ten octaves of added memory move the ratio by 15%. **Nonlinearity severity moves it by a factor of three, monotonically and on 8/8 seeds.** The hypothesis that motivated this section — memory length as the binding variable — is not supported. Sweeping τ over 5–150 UI at a harsher operating point reproduces the same flatness: the NN/table ratio stays within 1.21–2.66× with no trend in τ .

C. The real mechanism, and a defect in the comparison

Why should compression severity favour a network? It should not — by Proposition 1 a memoryless warp is exactly invertible by an ADC-code table at zero multipliers, and severity is precisely what that table absorbs. Inspecting the configuration revealed that the table path did not have one. The compressed signal was being fed directly to the NLMS front end, so as κ grew it was the *linear adapter* that degraded, not the table. Installing the memoryless inverse of Sec. V — which the paper already advocates, and which costs nothing — ahead of both structures repairs it:

TABLE XI

Effect of the zero-multiplier memoryless inverse ahead of the FFE. 25 dB, SNR 30 dB, 6.4×10^5 bits, 8 seeds, pooled BER.

Structure	$\kappa=1.0$	$\kappa=1.6$	$\kappa=2.0$
FFE 11 + DFE 8	2.42e-02	6.24e-02	8.25e-02
FFE 11 + table 2×4	3.92e-03	3.34e-02	5.87e-02
NN 5 → 2, 7-bit	3.48e-03	1.33e-02	1.88e-02
inv. + FFE 11 + table 2×4	1.06e-03	4.87e-03	1.13e-02
inv. + NN 5 → 2, 7-bit	1.64e-03	4.96e-03	6.36e-03

At $\kappa=2.0$ the table improves by 5.2× from a stage that costs no multipliers, and the apparent superiority of the network at $\kappa=1.6$ disappears. Neither structure is being measured until both are given the front end that Proposition 1 prescribes. **An architecture comparison is only as good as the weaker arm's configuration**, and for three editions the weaker arm was ours.

D. The crossover, measured fairly

With the memoryless inverse installed on both sides, the boundary is sharp and it is in κ , not in memory:

TABLE XII

Crossover in nonlinearity severity. Memoryless inverse ahead of both arms. 25 dB, SNR 30 dB, 6.4×10^5 bits, 8 seeds; ratios are per-seed medians.

κ	table / linear	NN / linear	NN / table	NN wins
1.0	3.48×	2.11×	0.59×	1/8
1.3	1.93×	1.41×	0.71×	0/8
1.6	1.84×	1.62×	0.78×	2/8
2.0	1.32×	2.23×	1.79×	8/8

The zero-multiplier arm holds to $\kappa \approx 1.6$ and loses at $\kappa=2.0$ on every seed. This is the regime that requires a network, and it is a regime of *severity*, not of memory. Adding droop to this sweep does not shift the boundary in a reproducible direction: the ratios become non-monotone in κ (0.81, 0.88, 1.44, 0.78), and we draw no conclusion from them.

E. Three attempts to rescue the table, all failed

Enlarge it. At every training budget a larger address space is worse, and the constraint is data rather than area: with 1.92×10^5 training symbols a 32 768-cell table populates 11 339 cells (35 %) against 256/256 for the small one, and its BER is six times worse. Preamble length is fixed by the standard, so this axis is closed to the designer.

Summarize the past into one address field. A leaky accumulator $m[n] = m[n-1] + (x[n] - m[n-1]) > k$ costs one shift and two adds and looks back 2^k samples; appending its code to the address should buy memory for a few bits instead of a few octaves. Measured, every configuration was worse than spending the same bits on one more raw sample ($4.20\text{e-}3$ and $5.38\text{e-}3$ against $2.87\text{e-}3$ for a 1024-cell two-sample table). An address bit is worth what it carries, and a neighbouring sample carried more than a slow mean.

Index on amplitude and past decisions jointly. Decisions are free (Sec. VI) and each costs one address bit, so a hybrid index is the natural union of the two structures of this paper — and, evaluated with *ideal* past decisions, it is an upper bound on any table-plus-DFE. It helps and it is not enough: $2.62\text{e-}3$ at 2048 cells against $3.63\text{e-}3$ for the plain table and $9.2\text{e-}4$ for the network on the same data. The ceiling of Proposition 3 is relaxed by the amplitude field but not removed.

F. Crosstalk, supply noise and timing error

These three are not equalizable by any of the structures considered — they are, respectively, another transmitter’s data, a time-varying gain and a displacement of the observation. The question they answer is whether the ordering survives them.

TABLE XIII

Robustness. Pooled BER, 4×10^5 bits, 4 seeds. Best in each row in bold.

Level	FFE 11	table 2×4	NN 5→2	MLP 21×16
FEXT 0.0	2.39e-02	2.24e-03	1.33e-03	9.05e-04
FEXT 0.1	2.47e-02	2.54e-03	1.59e-03	1.19e-03
FEXT 0.2	2.87e-02	4.76e-03	2.83e-03	2.34e-03
FEXT 0.3	3.60e-02	1.03e-02	6.03e-03	5.47e-03
RJ 0.05 UI	2.03e-02	1.48e-03	9.37e-04	6.10e-04
RJ 0.15 UI	3.11e-02	6.22e-03	3.69e-03	3.44e-03
supply ±25 %	2.30e-02	2.80e-03	1.70e-03	1.22e-03
supply ±50 %	2.16e-02	5.46e-03	2.36e-03	2.05e-03

The ordering is preserved throughout; no impairment reverses a conclusion of Sec. X. Two specific expectations were wrong. We expected a static table to be hurt more by a time-varying gain than a slicer is, because a stored cell acquires a different meaning at each phase of the modulation; measured over 4×10^5 bits and 4 seeds at ±50 %, the penalty is $2.43\times$ for the table against $2.66\times$ for the linear slicer — no asymmetry. And because $m < 1$ cannot change a sign, the absolute penalty from supply noise is small for every structure. Timing error is non-monotone in this regime (RJ 0.05 UI improves on RJ 0) because our interpolating sampler low-passes the waveform slightly; on a linear channel the same sweep degrades monotonically ($7.1\text{e-}5$ to $6.4\text{e-}4$ at 0.12 UI), so the non-monotonicity is an artifact of the impairment, not of the structures.

G. What the crossover costs

The network that wins at $\kappa=2.0$ is the same NN 5→2 of Table IX: 12 multipliers, 2336 LC, 47.87 MHz. It is *smaller and faster* than FFE 11 + table (3037 LC, 41.38 MHz). Where a network is genuinely required, it need not be a large one — and the multiplier-free thesis of this paper does not survive by an area argument at that point, it simply ceases to apply. Networks large enough to exploit a long observation, by contrast, are unbuildable on this device:

TABLE XIV

Place-and-route of wide networks, iCE40 HX8K (7680 LC), pipelined, Q2.4.

Design	Mults	LC	Utilization
NN 5→2	12	2336	30 %
NN 11→4	48	9515	123 % — does not fit
NN 21→4	88	17646	229 % — does not fit
NN 11→16	192	37371	486 % — does not fit

Area is very nearly proportional to multiplier count (198, 198, 201, 195 LC per multiplier), so the window length a network may use on a fixed device is set by that constant, not by the training procedure.

H. What is still missing

The channel of (32) is a physical parameterization, not a measurement. Obtaining Touchstone data requires network access that the environment in which this work was produced does not have, and we state this rather than presenting a model

as data. The gap is not cosmetic: with a via stub placed so that its notch falls inside the band ($\tau_s=0.6T$, notch at $0.72f_N$, -50 dB) every structure operates between 5 % and 14 % BER, a regime of information loss in which the equalizer choice is nearly irrelevant. With the notch moved just above the band ($\tau_s=0.35T$) the conclusions of Sec. X reappear intact: the table is best on the linear-plus-memoryless channel ($4.97\text{e-}4$ against $1.05\text{e-}3$ for the network, 0/6 network wins), and the network is best once a long-memory nonlinearity is added ($3.62\text{e-}3$ against $6.65\text{e-}3$, 6/6). Whether a measured channel lands nearer the first case or the second is the single largest open question in this work.

XII. MEASURED CHANNELS, DETECTION BOUNDS, AND DEVICE SCOPE

Three gaps remained: the channel was synthetic, the structures were compared only against each other, and the area comparison came from a single device with no hard multipliers. This section closes all three. Two of the three outcomes revise conclusions stated earlier in this paper.

A. Channels that are not ours

Two channels replace the fitted magnitude curve of Sec. II. The first is built from geometry and materials by an independent library (scikit-rf): a microstrip with the causal Djordjević–Svensson dielectric, Kirschning–Jansen dispersion, conductor resistivity and surface roughness, over which impedance discontinuities are cascaded as real line sections. Nothing in its magnitude or phase is chosen by us; we choose only w , h , t , ϵ_r , $\tan \delta$, ρ and length. At $200\text{ }\mu\text{m}/100\text{ }\mu\text{m}$, $\epsilon_r=3.7$, $\tan \delta=0.002$, 40 inch and 20 GBd it gives 26 dB at Nyquist, main cursor 0.22, and a closed eye.

The second is *measured*. scikit-rf ships a two-port vector-network-analyzer measurement (1–10 GHz, 91 points). One section is too mild to separate equalizers (0.98 dB at Nyquist), so we cascade K identical measured sections — a physically meaningful construction — and every decibel, ripple and reflection then comes from an instrument rather than from a model. We fix the sample rate at twice the highest measured frequency so the FFT grid never leaves the measured band; only the sub-1 GHz region is extrapolated to DC, which scikit-rf does causally. Extrapolating above 10 GHz to imitate 20 GBd would make the result our extrapolation, not a measurement, so we do not.

TABLE XV

Cascaded measured two-port. sps=4, 5 GBd, Nyquist 2.5 GHz — inside the measured band.

Sections K	Loss at Nyquist	Main cursor	ISI eye
1	0.98 dB	1.057	+2.08
8	10.61 dB	0.462	+0.38
16	19.65 dB	0.211	−0.23
24	28.77 dB	0.098	−0.40

B. What every structure is worth against the optimum

Until now this paper compared structures with each other. Such ratios have no floor: $7.98\times$ may be near the optimum or a thousand times from it, and the question “is a network

necessary” cannot be closed without knowing which. Three references settle it. The matched-filter bound over the oversampled waveform,

$$P_e \geq Q(\|p\|_2 / \sigma) \quad (33)$$

is the ISI-free limit of the waveform itself. Because every structure here observes one sample per symbol, the binding lower bound is the matched-filter bound over the symbol-spaced cursor vector $p_{\text{sym}}[k] = p[k_{\text{peak}} + kM]$:

$$P_e \geq Q(\|p_{\text{sym}}\|_2 / \sigma) \quad (34)$$

Both are calibrated rather than assumed: transmitting isolated pulses and detecting them with a matched filter reproduces (33) to 1.1–6.1 % with 234 and 967 errors counted, indistinguishable at 2σ . The third reference is *achievable*: a reduced-state sequence detector with per-survivor decision feedback (DDFSE), because the pulse response of a 25 dB channel spans more than sixty symbols and a full-memory Viterbi is not constructible — truncated at twelve symbols it plateaus at $2.8\text{e-}3$ even at 60 dB SNR, which is a model deficiency, not noise.

These references apply to the *nonlinear* link as well, by a data-processing argument we state as a proposition.

PROPOSITION 4. *Let the received waveform be $r = h * b + n$ and let every receiver impairment be a map Φ applied after n , with Φ not depending on b . Then for any detector $d(\Phi(r))$, $P_e \geq \min_d P_e(d'(r))$. If Φ is invertible, equality is attained by Φ^{-1} followed by the optimal detector for r .*

Proof. $b \rightarrow r \rightarrow \Phi(r)$ is a Markov chain, so the optimal detector on r is at least as good as any detector on $\Phi(r)$; invertibility makes the second link lossless. $\square$

The equality case is Proposition 1 extended from symbol-by-symbol estimation to sequence detection: a memoryless invertible front-end compression *costs an optimal receiver nothing*. We measure it. With the exact inverse installed and no quantizer, $\kappa=1.0$ reproduces the linear-channel BER of all three structures to the last digit ($6.46\text{e-}4$, $4.77\text{e-}4$, $9.71\text{e-}4$ in both cases), and the DDFSE makes zero errors in 150 000 symbols in both.

TABLE XVI

Structures against the optimum. 25 dB, SNR 30 dB, 6.4×10^5 bits, 4 seeds; DDFSE over 1.5×10^5 symbols. The reference detector sees unquantized samples, so the last three columns remove the ADC from the structures to separate that effect.

Impairment	MFB (34)	DDFSE	FFE+DFE	table	NN 5 → 2	FFE+DFE no ADC	table no ADC	NN no ADC
linear	1.1e-15	<2e-5	6.7e-4	5.0e-4	1.0e-3	6.5e-4	4.8e-4	9.7e-4
$\kappa=1.0$, memoryless	1.1e-15	<2e-5	3.4e-3	1.3e-3	1.4e-3	6.5e-4	4.8e-4	9.7e-4
$\kappa=2.0$, memoryless	1.1e-15	3.4e-3	1.4e-2	1.1e-2	7.3e-3	1.4e-3	9.4e-4	1.6e-3
$\kappa=1.0$, droop 18.8 UI	1.1e-15	6.7e-6	1.6e-3	1.1e-3	1.5e-3	1.2e-3	9.1e-4	1.3e-3
$\kappa=1.0$, $\beta=0.35$	1.1e-15	2.2e-2	1.1e-2	2.3e-3	4.2e-3	1.1e-3	2.2e-4	7.1e-4

Three things follow, and two of them are unfavourable to this paper.

Every structure leaves an order of magnitude on the table. On a linear channel, and on a channel with a memoryless nonlinearity, a sequence detector makes no errors where the best structure makes $5e-4$. The $3 \times$ – $8 \times$ differences this paper has been measuring are differences among detectors that are all $\geq 25 \times$ from what the same channel and noise permit. The gap is not caused by the nonlinearity — it is the price of deciding symbol by symbol.

The zero-multiplier inverse is not free once the ADC precedes it. At $\kappa=1.0$ the no-ADC columns are identical to the linear channel, but with a 7-bit ADC the same configuration degrades by $5.3 \times$ ($6.5e-4 \rightarrow 3.4e-3$). The arctanh stretch amplifies quantization noise in the saturated region. Sec. V

reported the inverse as costing nothing; it costs nothing in arithmetic, and it costs $5.3 \times$ in noise at 7 bits.

The β regime is the only one where the optimum is not a sequence detector. At $\beta=0.35$ the DDFSE reaches $2.2e-2$ while the 256-cell table reaches $2.3e-3$ — the table is ten times better than the best detector built on *any* linear response, including one whose response is estimated from the impaired waveform itself. This is the theoretical statement the paper has been missing: post-compression band-limiting produces a channel that no linear model describes, and that is exactly the condition under which a small nonlinear table earns its place. Everywhere else in the table, a linear-model sequence detector dominates both the table and the network.

C. The same comparison on the two new channels

TABLE XVII

Modelled line and measured cascade. 4×10^5 bits, 6 seeds, 7-bit ADC, memoryless inverse ahead of all arms. Ratios are per-seed medians.

Channel / impairment	DDFSE	FFE+DFE	table	NN 5 → 2	table/lin	NN/table	NN wins
line 40", $\kappa=1.0$	6.7e-5	6.8e-3	4.1e-3	7.3e-3	1.59×	0.57×	0/6
line 40", $\kappa=2.0$	5.0e-5	1.2e-2	1.2e-2	1.1e-2	1.04×	1.06×	4/6
line 40", $\beta=0.35$	2.5e-2	7.6e-3	2.2e-3	2.2e-3	3.43×	1.00×	3/6
measured $K=16$, $\kappa=1.0$	<2.5e-5	1.1e-2	9.3e-3	1.0e-2	1.18×	0.93×	1/6
measured $K=16$, $\kappa=2.0$	<2.5e-5	1.6e-2	1.3e-2	1.1e-2	1.26×	1.08×	5/6
measured $K=16$, $\beta=0.35$	2.8e-4	1.1e-2	1.3e-3	2.9e-3	8.48×	0.46×	0/6

The headline result survives the change of channel, and only in the regime it was derived for. On a measured channel with post-compression band-limiting the table gains $8.48 \times$ over the linear baseline — larger than the $7.98 \times$ of Table VII — and the network loses to it on 6/6 seeds. Away from that regime the table's advantage collapses to 1.04 – $1.59 \times$, which is the honest scope of the contribution: **the sample-indexed table is not a general-purpose equalizer, it is the matched structure for a nonlinearity with short memory.** The $\kappa=2.0$ crossover of Sec. XI also weakens on these channels ($1.06 \times$ and $1.08 \times$, 4/6 and 5/6), so the boundary located there should be read as specific to the synthetic channel.

D. Relation to LUT-based FFE taps

A reviewer points to a 28 Gb/s PAM-4 receiver in which the first three FFE taps are implemented as a lookup table, the entries being computed offline from ADC data and loaded into the table. The technique is classical distributed arithmetic: with x_k in two's complement,

$$\sum_k c_k x_k = -2^{B-1} R(a^{(B-1)}) + \sum_{b < B-1} 2^b R(a^{(b)}), \quad R[a] = \sum_k c_k a_k^{(35)}$$

where $a^{(b)}$ collects bit b of each tap input. We could not read the cited work — the environment in which this study was produced has no access to the publisher — so we take the description as given and compare against the technique rather than the paper.

The two tables store different kinds of object, and the difference is testable. $R[a]$ in (35) is a *linear* function of the address: its contents follow from the tap values by arithmetic, no training data is involved, and the computed result is bit-identical to a multiplier implementation. The table proposed here stores $E[b[n] \mid i[n]]$, which is not a linear function of the address and cannot be produced by any FIR. Fitting each table's contents by least squares against its own address bits separates them without argument: the distributed-arithmetic ROM is reproduced to a relative residual of $3e-15$, the trained decision table leaves 0.27. We also built (35) in RTL and verified against the multiplier FFE over 300 random stimuli:

not one bit differs. Distributed arithmetic changes the cost of a computation; the proposed table changes which function is computed.

On cost, the measurement is unfavourable to distributed arithmetic on this device, and the reason is instructive:

TABLE XVIII
FFE with the first N taps in distributed arithmetic, iCE40 HX8K, 50 MHz target, W7/DW7.

Design	Mults	LC	F_{max}
FFE 11, all multipliers	11	2237	52.38 MHz
FFE 11, $N=3$ in DA	8	2339	48.82 MHz
FFE 11, $N=4$ in DA	7	2701	49.56 MHz
FFE 11, $N=6$ in DA	5	5745	36.83 MHz

On an FPGA whose multipliers are already LUTs, replacing three of them with seven bit-plane ROM reads and a shift-add tree costs 102 LC and 3.6 MHz, and the cost grows as 2^N .

TABLE XIX
Synthesis cell counts for four families. “+nD” counts hard multiplier cells (ECP5 MULT18X18D, Xilinx DSP48E1), which are not included in the cell total. Synthesis only — see the caveat below.

Design	Mults	iCE40	ECP5	Xilinx 7	Intel ALM
FFE 11	11	2267	1575 +11D	406 +11D	483
FFE 11 + DFE 8	11	3290	2124 +11D	862 +11D	1029
FFE 11 + table 2×4	11	3053	2333 +11D	1316 +11D	1218
FFE 11, $N=3$ in DA (35)	8	2368	3238 +8D	1031 +8D	942
NN 5 → 2	12	2388	1240 +12D	391 +12D	485
NN 11 → 4	48	9612	6093 +48D	1202 +48D	1869
NN 11 → 16	192	37727	14529 +192D	4106 +192D	7256

On a device with hard multipliers the ordering inverts. On Xilinx 7-series the proposed table costs 1316 fabric cells while the network it was built to displace costs 391 — the network is $3.4\times$ smaller, and smaller than the 11-tap FFE alone, because its arithmetic leaves the fabric entirely for twelve DSP48 slices while the table’s cost is address decode and storage that no DSP absorbs. The same reversal holds on ECP5 and Intel ALM. Furthermore the wide network that cannot be built at all on the iCE40 (NN 11 → 16, 37 727 cells against 7680 available) is 4106 cells and 192 DSP48 on 7-series, which fits a mid-range part with room to spare.

The honest formulation of this paper’s cost claim is therefore narrower than “multiplier-free is cheaper”:

SCOPE OF THE AREA RESULT. Replacing multipliers with a lookup saves area only where a multiplier is itself built from the same fabric as the lookup. On the iCE40 that is true and the table wins. On any device with hard multipliers — every current FPGA family, and any ASIC where a 7×7 multiplier is a few hundred gates — the table’s fabric cost dominates and a small network is cheaper. The BER results of Sec. X and XII are unaffected; only the area argument is device-bound.

This also explains the distributed-arithmetic measurement of Sec. XII-D from the other direction: DA costs more on ECP5 (3238 cells against 1575) precisely because it moves work out of DSP slices and into fabric. The technique and our table are

Distributed arithmetic pays where a multiplier is genuinely expensive relative to a small ROM — a custom receiver front end, which is the setting of the cited work — and not here. This also bounds what our own result means: **the table of Sec. VIII is cheap on this device because it replaces a decision stage, not because tables are cheap in general.** The two techniques are composable: a receiver may compute the FFE partial sums by (35) and index the proposed decision table with the result, leaving $L-N$ multipliers in the entire data path.

E. Device scope beyond one FPGA

Every area figure in Table IX comes from a device with no hard multipliers, which is the least favourable setting for multiplier-based designs. Synthesizing the same RTL for families that do have DSP blocks reverses the conclusion, and we report it because it is the strongest objection to this paper’s thesis.

both bets on fabric being cheaper than multipliers, and that bet is won or lost by the device, not by the algorithm.

We report cell counts only. The only place-and-route tool available here targets iCE40; there is no ECP5, Xilinx or Intel router and no standard-cell library. No F_{max} , utilization, congestion, ASIC area, delay or power figure is claimed for any device other than the iCE40, and the reversal above should be read as an area-proxy result awaiting place-and-route confirmation. Quoting a frequency from synthesis alone would be exactly the kind of unverified green light this work is otherwise at pains to avoid.

XIII. DISCUSSION AND LIMITATIONS

The comparisons in Sec. V–XI are among detectors that are all far from optimal. Sec. XII-B measures it: on a linear channel and under a memoryless nonlinearity a sequence detector makes no errors where the best structure here makes $5e-4$. The one regime in which that is not so is $\beta=0.35$, where no linear response describes the channel and the 256-cell table beats the sequence detector tenfold. The scope of this paper’s contribution is that regime, and readers should not carry its ratios outside it.

The impairment model determines the conclusion. This is the principal threat to validity. Section V shows that a memoryless warp is best undone by a 128-entry table; Section VIII shows that a memory-2 warp is best undone by a

256-entry table. Both conclusions are statements about the *structure* of the impairment. Sec. XI constructs the case in which the zero-multiplier arm loses, and locates it in the *severity* of the nonlinearity ($\kappa \approx 2$) rather than in its *memory*: ten octaves of added memory move the comparison by 15 %, while severity moves it by a factor of three. The address-space argument that motivated this paper is therefore correct in form and wrong about which variable binds first.

Channel and device scope. Sec. XII-A replaces the fitted channel with one built from geometry and materials by an independent library, and with a cascade of a measured two-port; the headline result survives on both, and only inside the β regime. What remains missing is a measured *backplane*: the instrument data we could obtain covers 1–10 GHz and a mild interconnect, so the hard channel is built by cascading it. Area beyond the iCE40 is reported as synthesis cell counts only (Sec. XII-E); there is no router for other families here and no standard-cell library, so no $F_{\max}$ or ASIC figure is claimed. The target device has no DSP blocks, so all multipliers map to LUTs and the relative cost of a multiplier is at its most unfavourable; on a DSP-bearing FPGA or an ASIC the area comparison would shift in favour of the multiplier-based designs.

NRZ only. Two results depend on binary decisions: DFE taps are multiplier-free because $d \in \{\pm 1\}$, and the table stores one bit per cell. Under PAM4 neither holds, the decision alphabet grows, and the address space of any sample-indexed table grows with it.

Timing recovery. Sec. XI-F adds residual random and sinusoidal sampling jitter (31) and finds the ordering unchanged to 0.15 UI rms, but no CDR loop is simulated: the model applies jitter that a loop has already failed to track and cannot reproduce loop bandwidth, pattern-dependent jitter, or the interaction between a nonlinear detector and the phase detector that feeds it. That interaction is untested.

Rare-event behaviour. The operating BERs here are 10^{-2} – 10^{-3} . Counting to a compliance floor of 10^{-12} would require 3×10^{12} bits — 242 core-days at the measured 1.4×10^5 bit/s — so statistical extrapolation is unavoidable, and the usual Gaussian-tail argument does not transfer cleanly to a nonlinear decision boundary. A table has a specific exposure here: underpopulated cells emit a default, and nothing guarantees that default is correct for rare excursions. This is untested and is the most likely place for the present results to fail.

Adaptation protocol. Standards prescribe how equalizer coefficients are negotiated, typically as increment/decrement handshakes over a back channel. Neither a weight matrix nor a populated table speaks that language. The table is arguably closer — it is filled by one counting pass over the preamble, so its convergence criterion reduces to samples-per-cell — but mapping either onto a standard link-training sequence is unaddressed here.

XIV. CONCLUSION

The common motivation for neural equalization in wireline receivers — that linear equalizers cannot invert front-end nonlinearity — is true but insufficient. For a memoryless

nonlinearity the optimal memoryless corrector is a 2^B -entry table with no multipliers, and measured on our model it outperforms a 96-multiplier MLP by 2.02 $\times$, after which the network is redundant.

Where nonlinearity does have memory, we showed that all structures observing only past decisions — linear DFE, decision-indexed ROM, and multiplier-free Volterra expansions — saturate near 3.1 $\times$ irrespective of parameter count, because decisions discard the amplitude that the impairment mixes. Changing the observation rather than the parameter budget removes the ceiling: a 256-entry, one-bit-per-entry table indexed by two quantized FFE output samples achieves 7.98 $\times$ with zero back-end multipliers, exceeding a 96-multiplier network that does not fit the target device, at +1.3 % area and 1.19 $\times$ the clock of the linear baseline.

The organizing principle is that impairments decompose and each class has a matched corrector with a characteristic hardware cost: memoryless warp $\rightarrow$ code table (0 multipliers); linear pre-cursor ISI $\rightarrow$ FFE (1 multiplier per tap); linear post-cursor ISI $\rightarrow$ DFE (0); decision-dependent nonlinearity $\rightarrow$ decision table (0); sample-dependent nonlinearity with memory $\rightarrow$ sample table (0) up to the address space it can populate, and only beyond that a parametric network. A general-purpose MLP is expensive precisely because it does not exploit this decomposition, spending multipliers on work that costs none in the matched structure.

Two limits of that statement were then measured rather than assumed, and both narrow it. *In detection terms*, all of these structures — ours included — are at least 25 $\times$ from what the channel and noise permit, because they decide symbol by symbol; the sole exception is post-compression band-limiting, where no linear response describes the channel and the 256-entry table beats the best linear-model sequence detector tenfold. That regime, and not nonlinearity in general, is where the contribution lives, and it survives on a channel built from geometry and materials by an independent library and on a cascade of a measured two-port (8.48 $\times$ there, against 1.04–1.59 $\times$ outside it). *In hardware terms*, “multiplier-free” is a statement about a device class: on the iCE40 the table is cheaper, but on any family with hard multipliers the same network costs 391 fabric cells against the table’s 1316, because the table’s cost is address decode and storage that no DSP absorbs. The decomposition above remains the useful organizing principle; what it buys depends on which fabric the matched corrector is spent on.

APPENDIX: REPRODUCTION PARAMETERS

Unless stated otherwise: $L_{\text{dB}} = 25$ dB, $\text{SNR} = 30$ dB, $M = 8$ samples/symbol, $\kappa = 1.0$, $\beta = 0.35/T$ where a post-filter is used, $B = 7$ -bit ADC at $\rho = 2.5\sigma$, FFE 11 taps at 7-bit coefficients and 7-bit data, DFE 8 taps at 8 bits, 3×10^5 symbols per run with the first 30 % used for adaptation and the remainder for error counting, and 6–8 independent seeds per point. Neural equalizers use *hardtanh*, Adam, 10–12 epochs, and Q2.4 fixed point in 7 bits. Table training uses $N_{\min} = 4$. All reported multi-seed comparisons are means of per-seed ratios, with the observed minimum and maximum quoted as the spread.